Diary Data Analysis

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Contents

Exploratory: Impact of CU on training **3**

1) Child rated ASB at 2 weeks - by CU **3**

1.1) Table of outcomes . . . 3

1.3) Plot of outcomes . . . 4

1.3) Linear Regression models - in line with other analyses . . . 6

1.3.1) Does interaction add anything to model? - No . . . 6

1.3.2) LM Results: . . . 6

1.4) Bootstrap linear regression model (in line with other analyses) . . . 8

1.4.1) Results and specification . . . 8

1.4.2) Confidence intervals . . . 8

1.4.3) Model diagnostics . . . 10

2) Teacher rated ASB at two weeks **13**

2.1) Table of outcomes . . . 13

2.2) plots of outcomes . . . 14

2.3) Linear Regression model . . . 16

2.3.1) Is model with interaction better than model without? - No . . . 16

2.3.2) Model results . . . 16

2.3.3) Assumption checks . . . 17

2.4) Bootstrapped linear regression . . . 18

2.4.1) Results and specification . . . 18

2.4.2) Confidence Intervals . . . 18

2.4.3) Model Diagnostics . . . 19

3) Child Prosocial at two weeks **24**

3.1) Table of outcomes . . . 25

3.2) Plots of outcomes . . . 26

3.3) Linear Regression . . . 27

3.3.1) is model with interaction better fit? No . . . 27

3.3.2) Model summary . . . 28

3.3.3) Assumption checks . . . 29

3.4) Bootstrapped linear regression . . . 30

3.4.1) Results . . . 30

3.4.2) Confidence Intervals . . . 30

3.4.3) Model diagnostics . . . 32

4) Teacher Prosocial at 2 weeks **35**

4.1) Table of outcomes . . . 35

4.2) Plots of outcomes . . . 36

4.3) Linear Regression model . . . 38

4.3.1) Does interaction term improve model? No . . . 38

4.3.2) Model results . . . 38

4.3.3) Assumption checks . . . 39

4.4) Bootstrapped model . . . 39

4.4.1) Results and specification . . . 39

4.4.2) Confidence Intervals . . . 40

4.4.5) Model diagnostics . . . 42

Exploratory: Impact of CU on training

1) Child rated ASB at 2 weeks - by CU

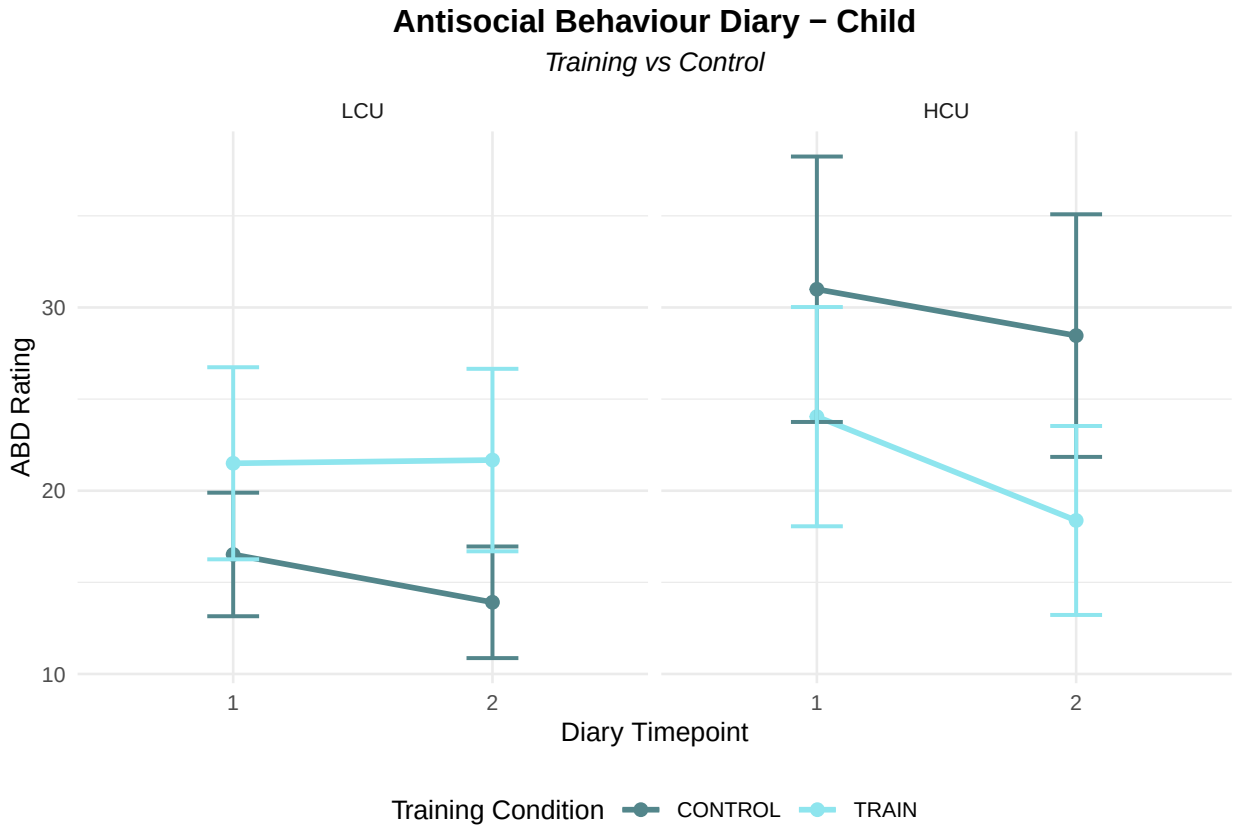
1.1) Table of outcomes

Table of outcomes:

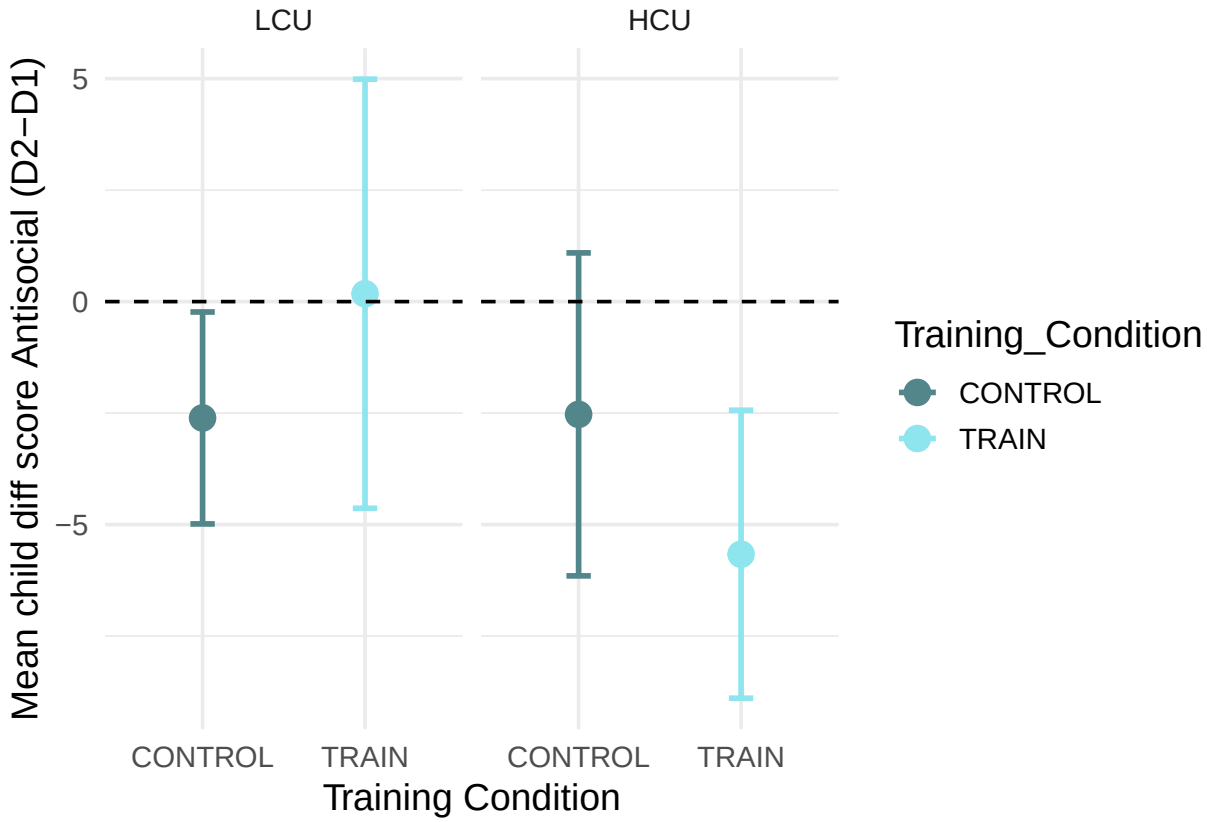
Training_Condition	diaryTP	CP_Group	Group_N_ABD	Mean_child_rated_ABD	se_child_rated_ABD	NA_child_rated_ABD
CONTROL	1	LCU	26	16.52077	3.370843	0
CONTROL	1	HCU	18	30.99333	7.242612	0
CONTROL	2	LCU	26	13.91154	3.045668	0
CONTROL	2	HCU	18	28.46444	6.619444	0
TRAIN	1	LCU	22	21.49636	5.240413	0
TRAIN	1	HCU	16	24.04125	5.982017	0
TRAIN	2	LCU	22	21.67318	4.978893	0
TRAIN	2	HCU	16	18.37625	5.155261	0

1.3) Plot of outcomes

First plotting data:



Difference plot:



1.3) Linear Regression models - in line with other analyses

```
# simple linear model (in line with other analyses)
model_c_asb_CU_lm <- lm(ASB_sum_D2 ~ ASB_sum_D1 + Training_Condition + CP_Group +
                         Training_Condition*CP_Group,
                       data =  child_asb_d1d2_wide)

# model without interaction
model_c_asb_CU_lm_nointeraction <- lm(ASB_sum_D2 ~ ASB_sum_D1 + Training_Condition + CP_Group,
                       data =  child_asb_d1d2_wide)
```

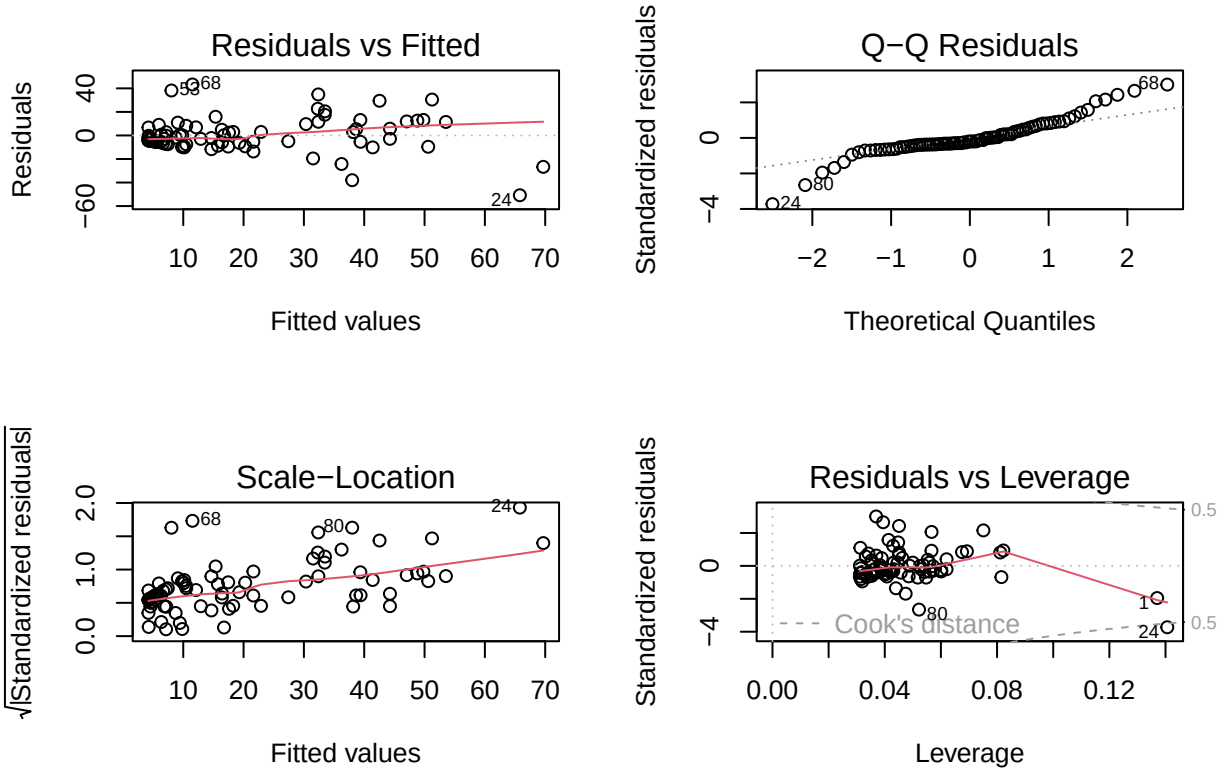
1.3.1) Does interaction add anything to model? - No

```
## Analysis of Variance Table
##
## Model 1: ASB_sum_D2 ~ ASB_sum_D1 + Training_Condition + CP_Group
## Model 2: ASB_sum_D2 ~ ASB_sum_D1 + Training_Condition + CP_Group + Training_Condition *
##     CP_Group
##   Res.Df   RSS Df Sum of Sq     F Pr(>F)
## 1     78 16854
## 2     77 16396  1    458.66 2.154 0.1463
```

1.3.2) LM Results:

```
##
## Call:
## lm(formula = ASB_sum_D2 ~ ASB_sum_D1 + Training_Condition + CP_Group,
##     data = child_asb_d1d2_wide)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -50.801  -5.818  -2.901   6.513  43.306
##
## Coefficients:
##                          Estimate Std. Error t value Pr(>|t|)
## (Intercept)               4.23401    2.89971   1.460    0.148
## ASB_sum_D1                0.69586    0.06898  10.088 8.53e-16 ***
## Training_ConditionTRAIN   0.33151    3.25558   0.102    0.919
## CP_GroupHCU               0.03662    3.35219   0.011    0.991
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 14.7 on 78 degrees of freedom
## Multiple R-squared:  0.5747, Adjusted R-squared:  0.5583
## F-statistic: 35.13 on 3 and 78 DF,  p-value: 1.805e-14
```

1.3.3) LM Assumption Checks



1.4) Bootstrap linear regression model (in line with other analyses)

1.4.1) Results and specification

```
boot_coef_c_asb <- function(data, indices) {
  d <- data[indices, ]  # Resample with replacement
  fit <- lm(ASB_sum_D2 ~ ASB_sum_D1 + Training_Condition + CP_Group,
                    data =  d)
  return(coef(fit))
}

# Run bootstrap
## Boot function: Generates R bootstrap replicates of a statistic applied to data.
## data = my data
## statistic = A function which, when applied to data returns a vector containing the statistic(s) of i
set.seed(123)  # For reproducibility
boot_results_c_asb <- boot(data = child_asb_d1d2_wide,
                      statistic = boot_coef_c_asb,
                      R = 5000)

# View results
boot_results_c_asb
```

```
##
## ORDINARY NONPARAMETRIC BOOTSTRAP
##
##
## Call:
## boot(data = child_asb_d1d2_wide, statistic = boot_coef_c_asb,
##     R = 5000)
##
##
## Bootstrap Statistics :
##       original      bias    std. error
## t1* 4.23400670 -0.14208511   2.2964985
## t2* 0.69585839  0.01036699   0.1062558
## t3* 0.33151420  0.18244419   3.3052723
## t4* 0.03662278 -0.20235615   3.2882893
```

1.4.2) Confidence intervals

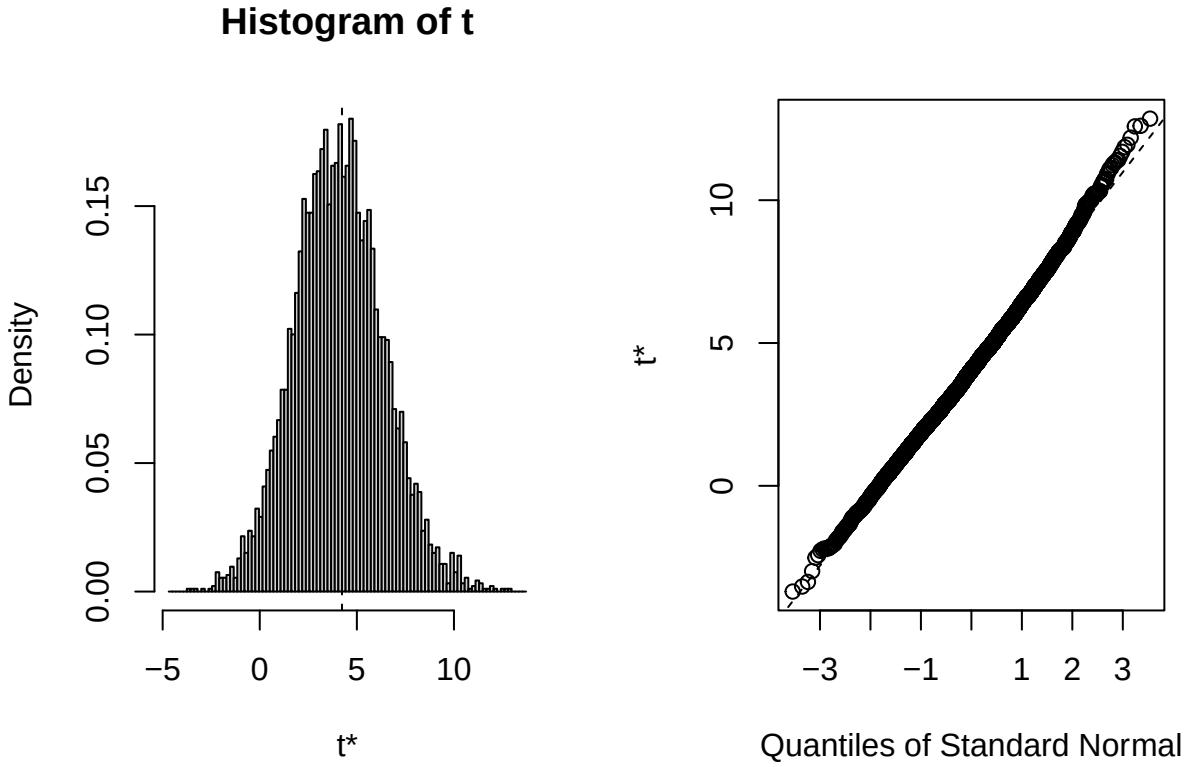
```
## BOOTSTRAP CONFIDENCE INTERVAL CALCULATIONS
## Based on 5000 bootstrap replicates
##
## CALL :
## boot.ci(boot.out = boot_results_c_asb, type = "bca", index = 1)
##
## Intervals :
## Level       BCa
## 95%   ( 0.318,  9.794 )
## Calculations and Intervals on Original Scale
```

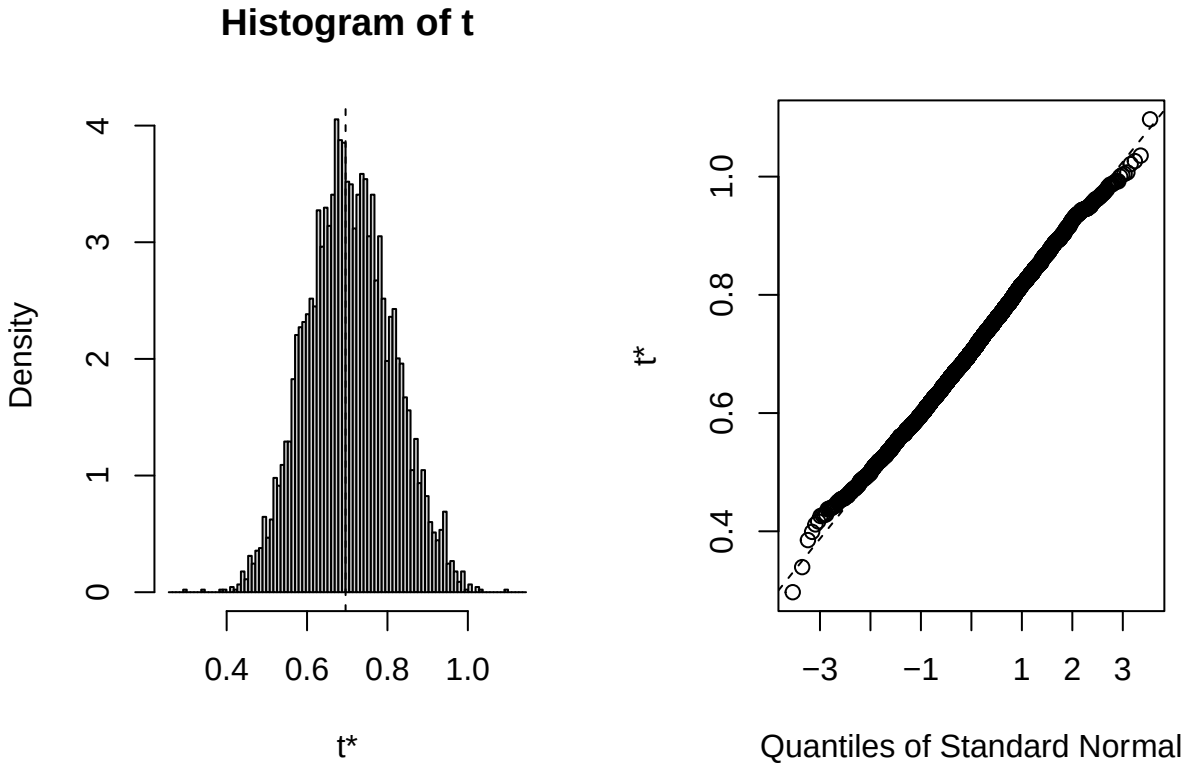
```
## BOOTSTRAP CONFIDENCE INTERVAL CALCULATIONS
## Based on 5000 bootstrap replicates
##
## CALL :
## boot.ci(boot.out = boot_results_c_asb, type = "bca", index = 2)
##
## Intervals :
## Level       BCa
## 95%   ( 0.4645,  0.8835 )
## Calculations and Intervals on Original Scale

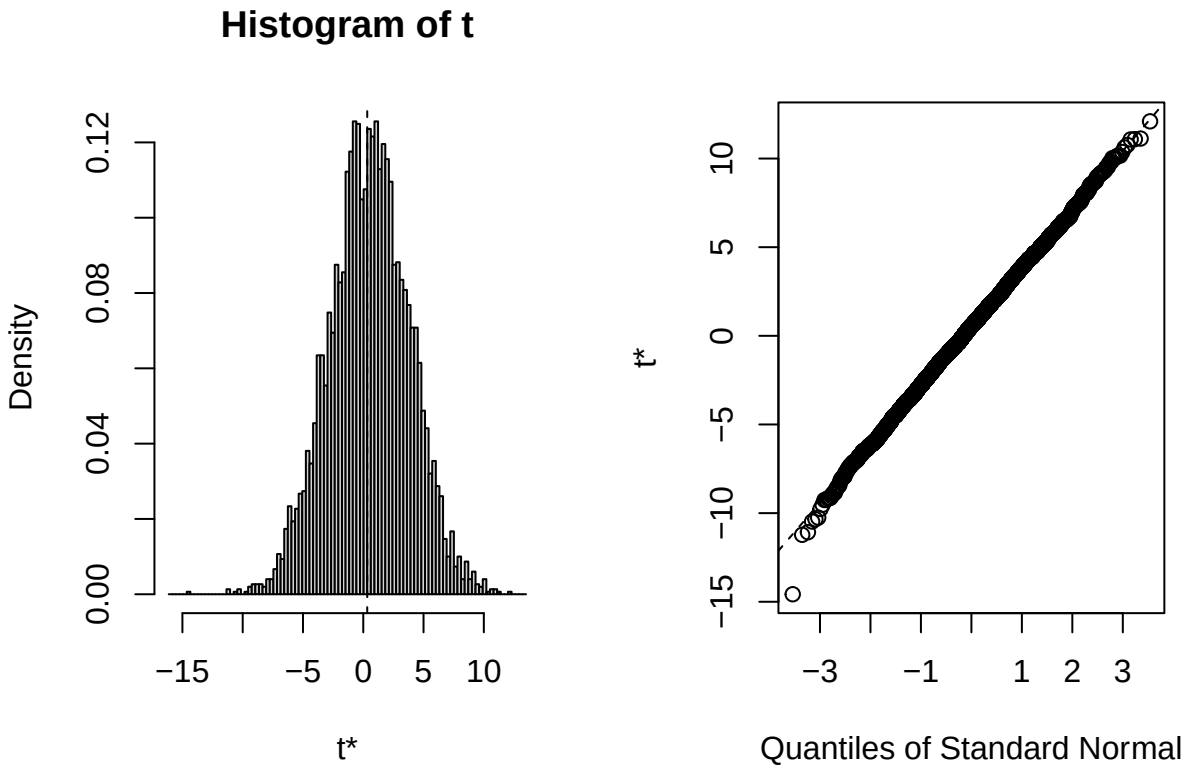
## BOOTSTRAP CONFIDENCE INTERVAL CALCULATIONS
## Based on 5000 bootstrap replicates
##
## CALL :
## boot.ci(boot.out = boot_results_c_asb, type = "bca", index = 3)
##
## Intervals :
## Level       BCa
## 95%   (-6.4196,  6.4558 )
## Calculations and Intervals on Original Scale

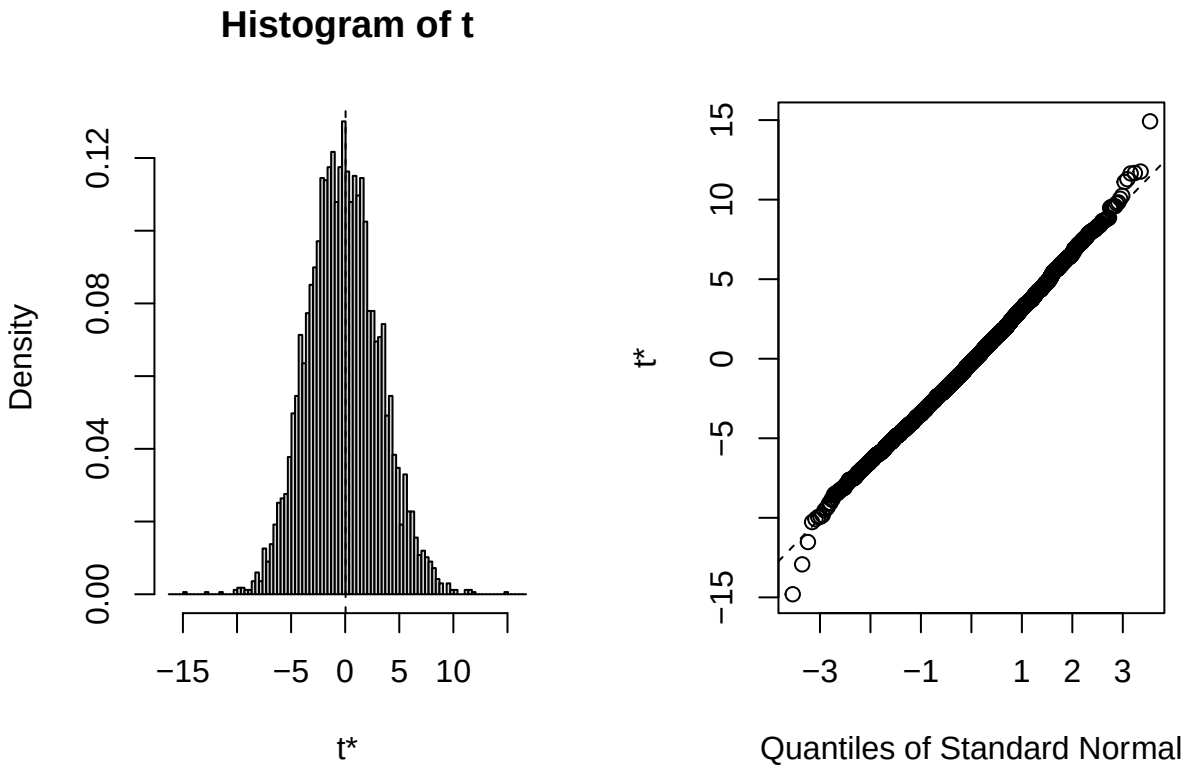
## BOOTSTRAP CONFIDENCE INTERVAL CALCULATIONS
## Based on 5000 bootstrap replicates
##
## CALL :
## boot.ci(boot.out = boot_results_c_asb, type = "bca", index = 4)
##
## Intervals :
## Level       BCa
## 95%   (-5.6906,  7.4275 )
## Calculations and Intervals on Original Scale
```

1.4.3) Model diagnostics






Histogram of t
Density
0.00
0.04
0.08
0.12
−15
−5
0
5
10
t*
t*
−15
−10
−5
0
5
10
−3
−1
1
2
3
Quantiles of Standard Normal


Histogram of t
Density
0.00
0.04
0.08
0.12
−15
−5
0
5
10
t*
t*
−15
−5
0
5
10
15
−3
−1
1
2
3
Quantiles of Standard Normal

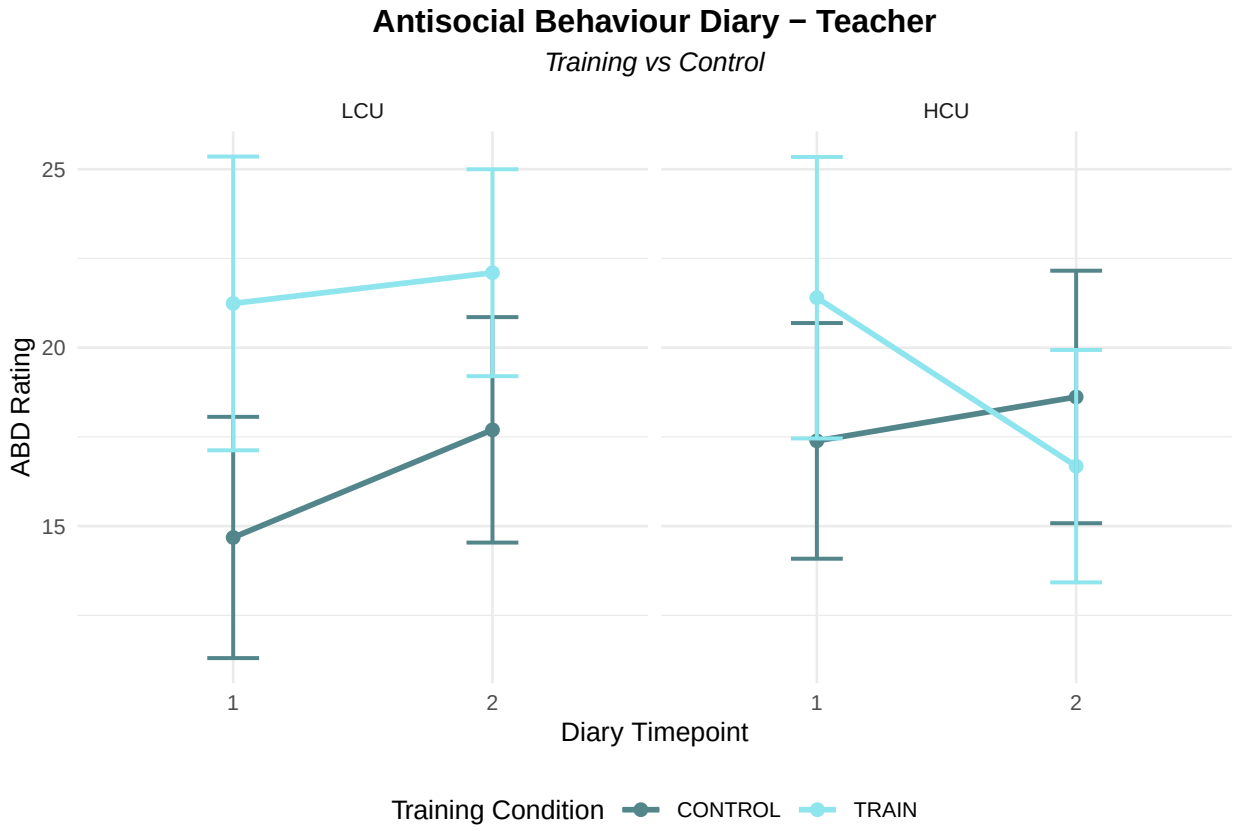
2) Teacher rated ASB at two weeks

2.1) Table of outcomes

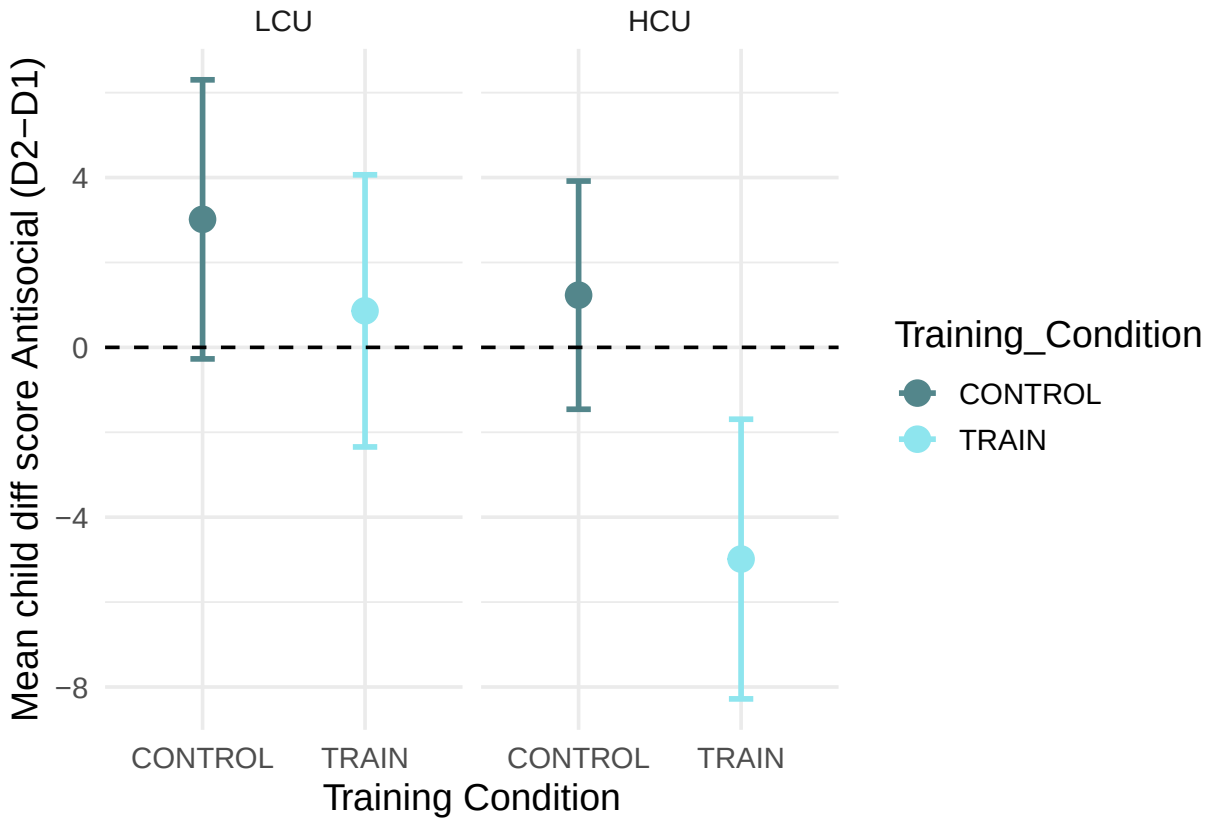
Training_Condition	diaryTP	CP_Group	Group_N	Mean_ABD	se_ABD	Mean_PBD	se_PBD
CONTROL	1	LCU	32	14.68229	3.379629	37.48958	4.023299
CONTROL	1	HCU	21	17.38889	3.301882	18.07937	3.305231
CONTROL	2	LCU	32	17.69792	3.158693	35.24479	3.616988
CONTROL	2	HCU	21	18.61905	3.537556	24.12698	4.220790
TRAIN	1	LCU	25	21.24000	4.115801	39.55333	4.566927
TRAIN	1	HCU	25	21.40000	3.943220	27.64000	3.650338
TRAIN	2	LCU	25	22.10000	2.898211	38.41333	4.598401
TRAIN	2	HCU	25	16.68000	3.256161	30.14667	3.809410

2.2) plots of outcomes

First plotting data:



Difference plot:


LCU
HCU
Mean child diff score Antisocial (D2−D1)
4
0
−4
−8
CONTROL
TRAIN
CONTROL
TRAIN
Training Condition
Training_Condition
CONTROL
TRAIN

2.3) Linear Regression model

```
lm_model_t_asb_CU <- lm(ASB_mean_D2 ~ ASB_mean_D1 + Training_Condition + CP_Group +
                         Training_Condition*CP_Group,
                       data =  teach_asb_d1d2_wide)


# model without interaction
lm_model_t_asb_CU_nointeraction <- lm(ASB_mean_D2 ~ ASB_mean_D1 + Training_Condition + CP_Group,
                       data =  teach_asb_d1d2_wide)
```

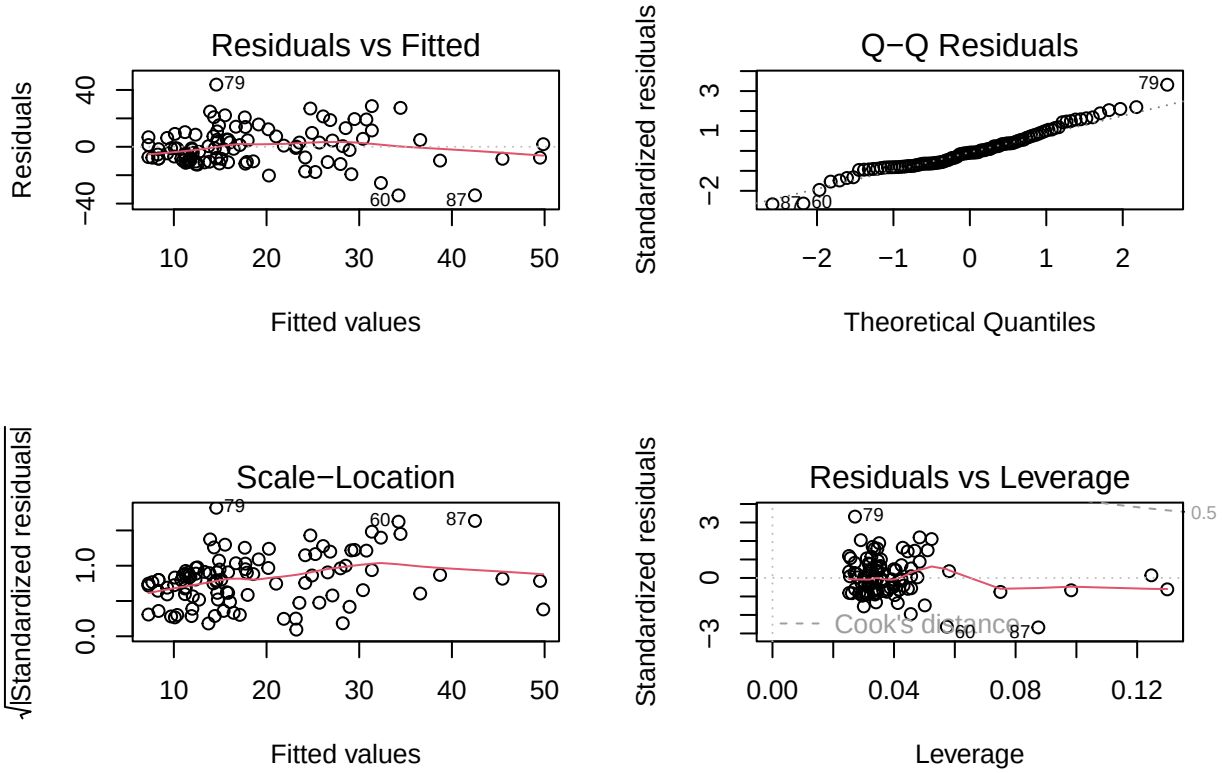
2.3.1) Is model with interaction better than model without? - No

```
## Analysis of Variance Table
##
## Model 1: ASB_mean_D2 ~ ASB_mean_D1 + Training_Condition + CP_Group + Training_Condition *
##     CP_Group
## Model 2: ASB_mean_D2 ~ ASB_mean_D1 + Training_Condition + CP_Group
##   Res.Df   RSS Df Sum of Sq     F Pr(>F)
## 1     97 17378
## 2     98 17526 -1   -148.16 0.827 0.3654
```

2.3.2) Model results

```
##
## Call:
## lm(formula = ASB_mean_D2 ~ ASB_mean_D1 + Training_Condition +
##     CP_Group, data = teach_asb_d1d2_wide)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -34.238  -8.983  -1.141   7.159  43.754
##
## Coefficients:
##                          Estimate Std. Error t value Pr(>|t|)
## (Intercept)              11.19668    2.37760   4.709 8.20e-06 ***
## ASB_mean_D1               0.50744    0.07154   7.093 2.06e-10 ***
## Training_ConditionTRAIN -1.08956    2.69451  -0.404    0.687
## CP_GroupHCU             -2.84772    2.68209  -1.062    0.291
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 13.37 on 98 degrees of freedom
## Multiple R-squared:  0.3432, Adjusted R-squared:  0.3231
## F-statistic: 17.07 on 3 and 98 DF,  p-value: 5.372e-09
```

2.3.3) Assumption checks



2.4) Bootstrapped linear regression

2.4.1) Results and specification

```
##
## ORDINARY NONPARAMETRIC BOOTSTRAP
##
##
## Call:
## boot(data = teach_asb_d1d2_wide, statistic = boot_coef_t_asb,
##     R = 5000)
##
##
## Bootstrap Statistics :
##      original       bias    std. error
## t1* 11.196675 -0.006794216  2.45059600
## t2*  0.507439  0.004445028  0.08658768
## t3* -1.089563 -0.053788453  2.71050626
## t4* -2.847717 -0.039839280  2.57091507
```

2.4.2) Confidence Intervals

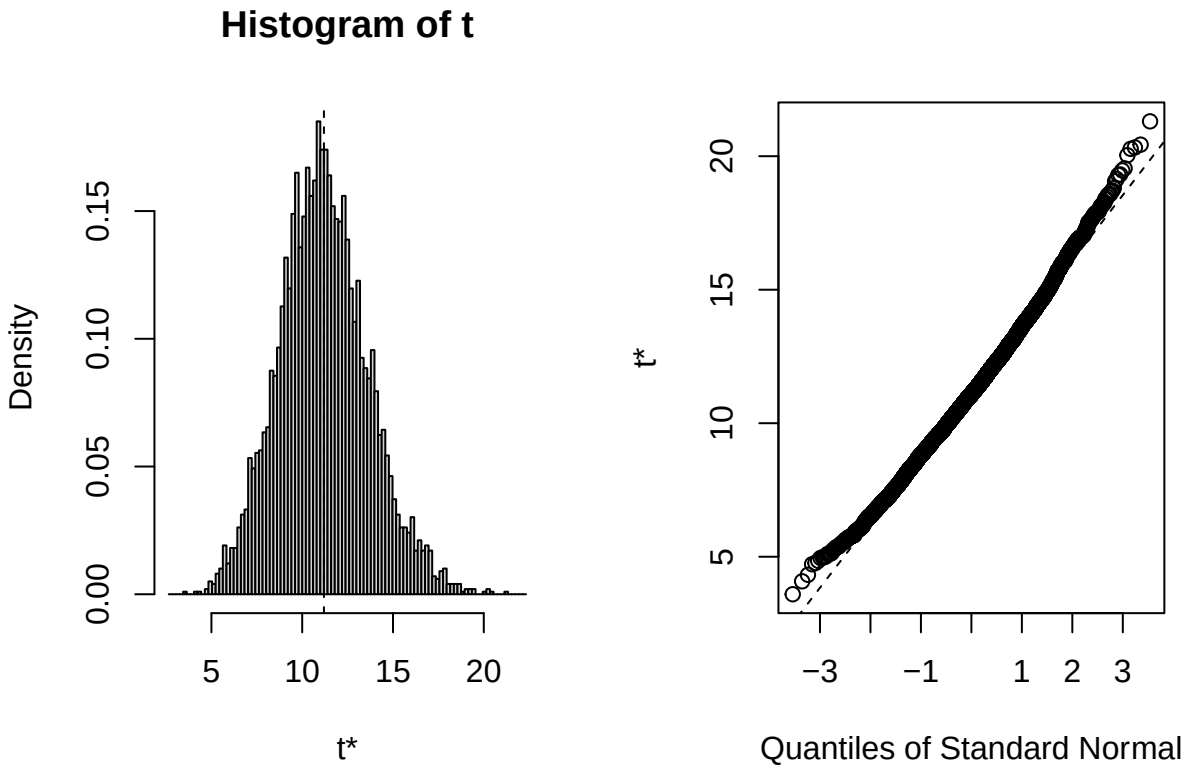
```
## BOOTSTRAP CONFIDENCE INTERVAL CALCULATIONS
## Based on 5000 bootstrap replicates
##
## CALL :
## boot.ci(boot.out = boot_results_t_asb, type = "bca", index = 1)
##
## Intervals :
## Level       BCa
## 95%   ( 7.03, 16.97 )
## Calculations and Intervals on Original Scale

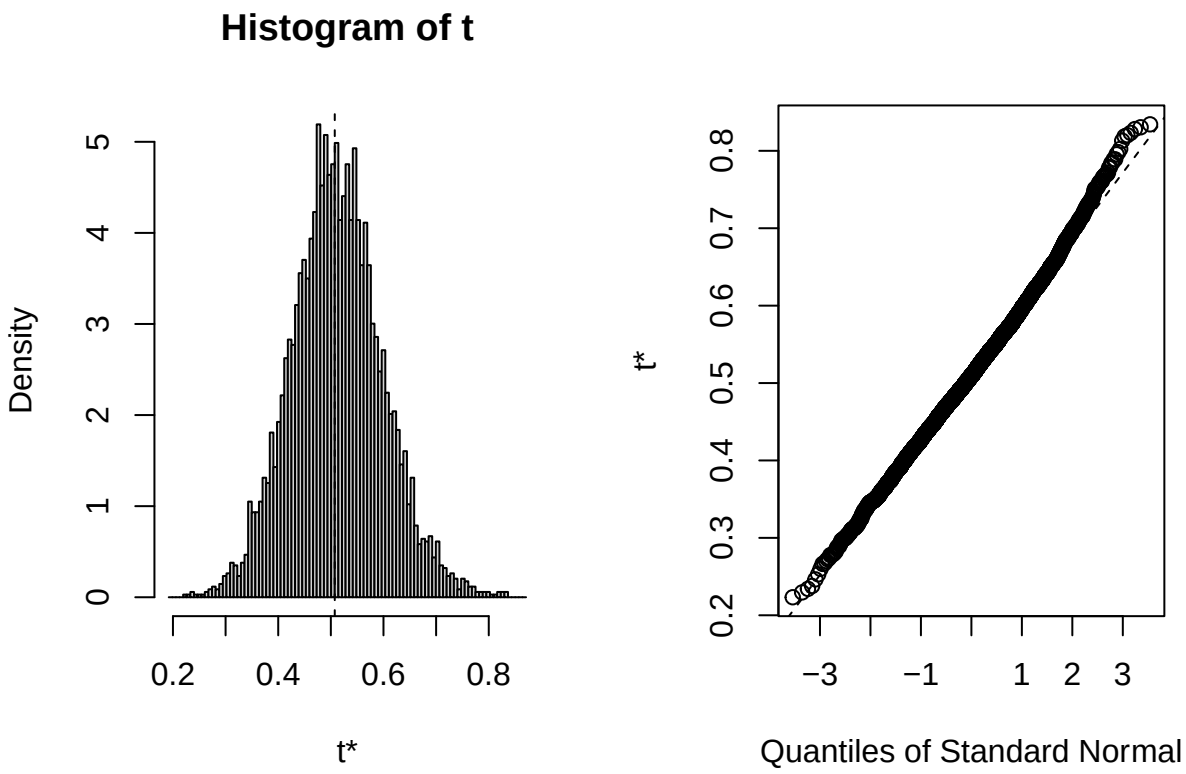
## BOOTSTRAP CONFIDENCE INTERVAL CALCULATIONS
## Based on 5000 bootstrap replicates
##
## CALL :
## boot.ci(boot.out = boot_results_t_asb, type = "bca", index = 2)
##
## Intervals :
## Level       BCa
## 95%   ( 0.3317,  0.6741 )
## Calculations and Intervals on Original Scale

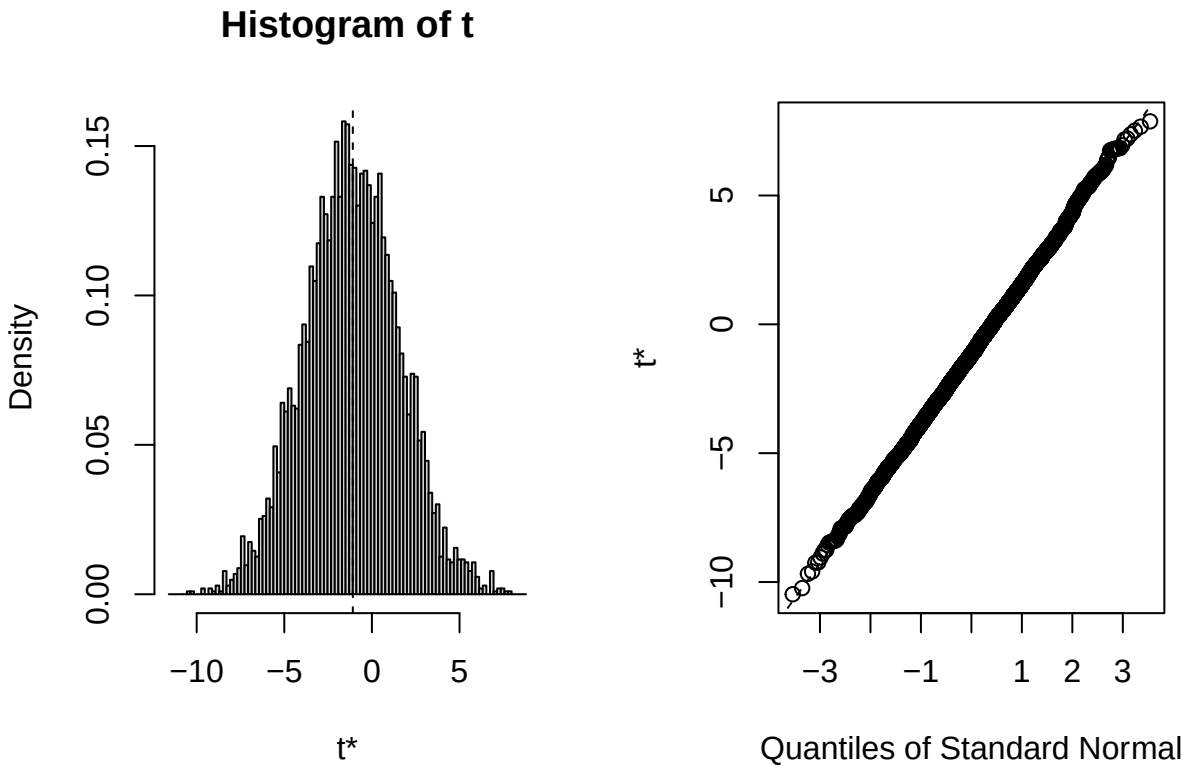
## BOOTSTRAP CONFIDENCE INTERVAL CALCULATIONS
## Based on 5000 bootstrap replicates
##
## CALL :
## boot.ci(boot.out = boot_results_t_asb, type = "bca", index = 3)
##
## Intervals :
## Level       BCa
## 95%   (-6.270,  4.461 )
## Calculations and Intervals on Original Scale
```

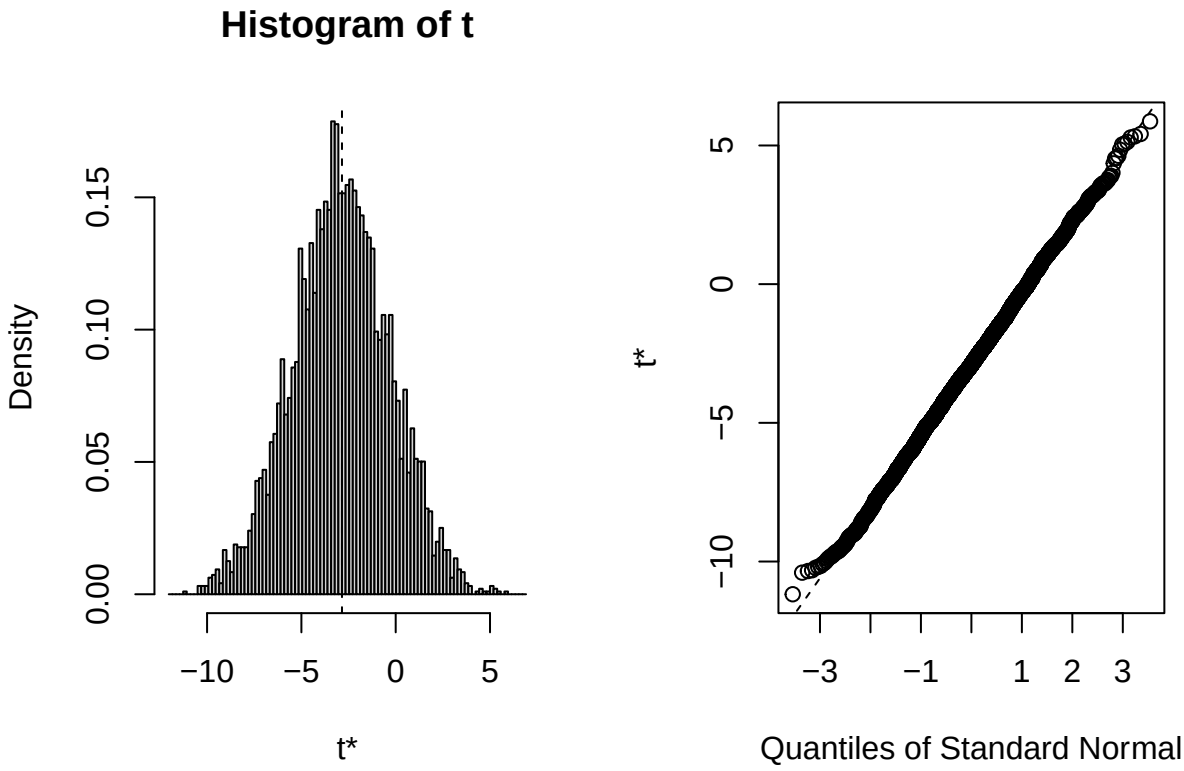
```
## BOOTSTRAP CONFIDENCE INTERVAL CALCULATIONS
## Based on 5000 bootstrap replicates
##
## CALL :
## boot.ci(boot.out = boot_results_t_asb, type = "bca", index = 4)
##
## Intervals :
## Level       BCa
## 95%   (-7.895,  2.269 )
## Calculations and Intervals on Original Scale
```

2.4.3) Model Diagnostics


Histogram of t
Density
0.00
0.05
0.10
0.15
5
10
15
20
t*
t*
5
10
15
20
−3
−1
1
2
3
Quantiles of Standard Normal


Histogram of t
Density
0
1
2
3
4
5
0.2
0.4
0.6
0.8
t*
t*
0.2
0.3
0.4
0.5
0.6
0.7
0.8
−3
−1
1
2
3
Quantiles of Standard Normal


Histogram of t
Density
0.00
0.05
0.10
0.15
−10
−5
0
5
t*
t*
−10
−5
0
5
−3
−1
1
2
3
Quantiles of Standard Normal


Histogram of t
Density
0.00
0.05
0.10
0.15
−10
−5
0
5
t*
t*
5
0
−5
−10
−3
−1
1
2
3
Quantiles of Standard Normal

3) Child Prosocial at two weeks

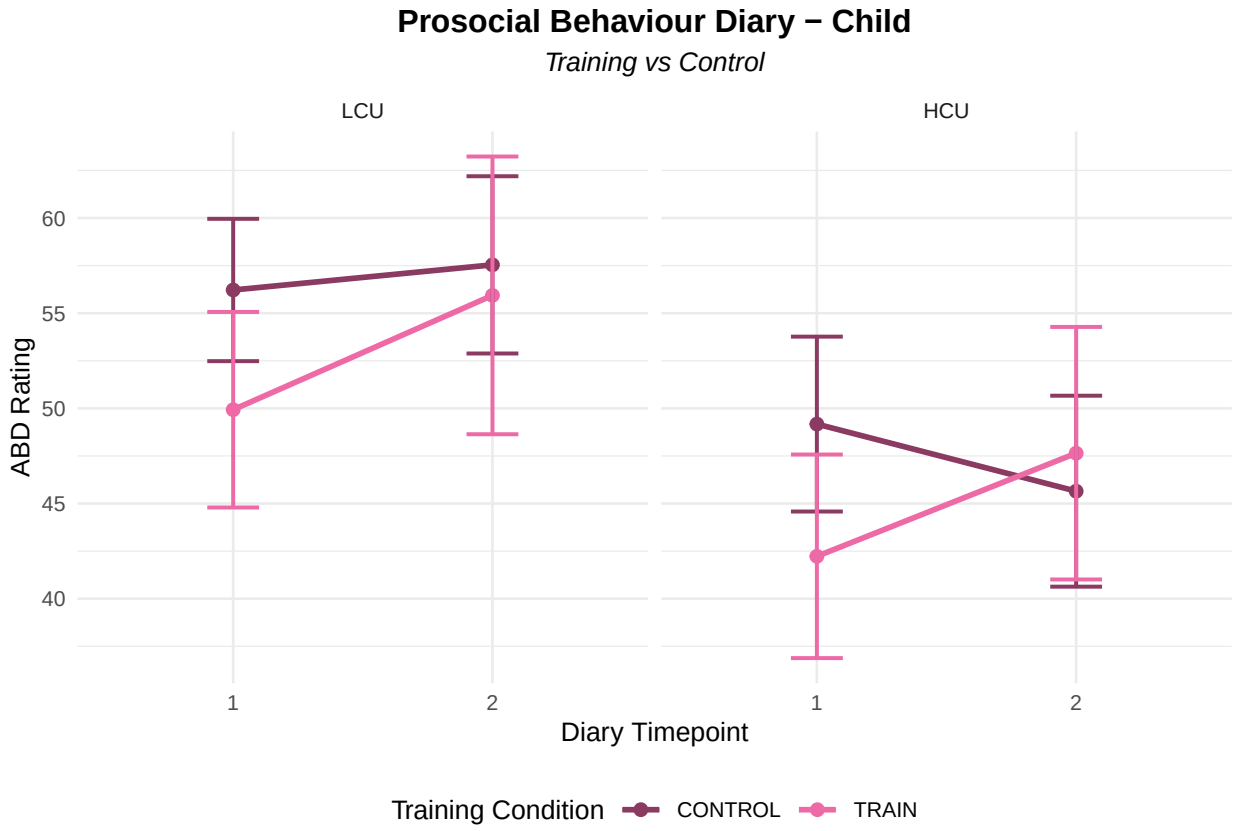
3.1) Table of outcomes

Table of outcomes:

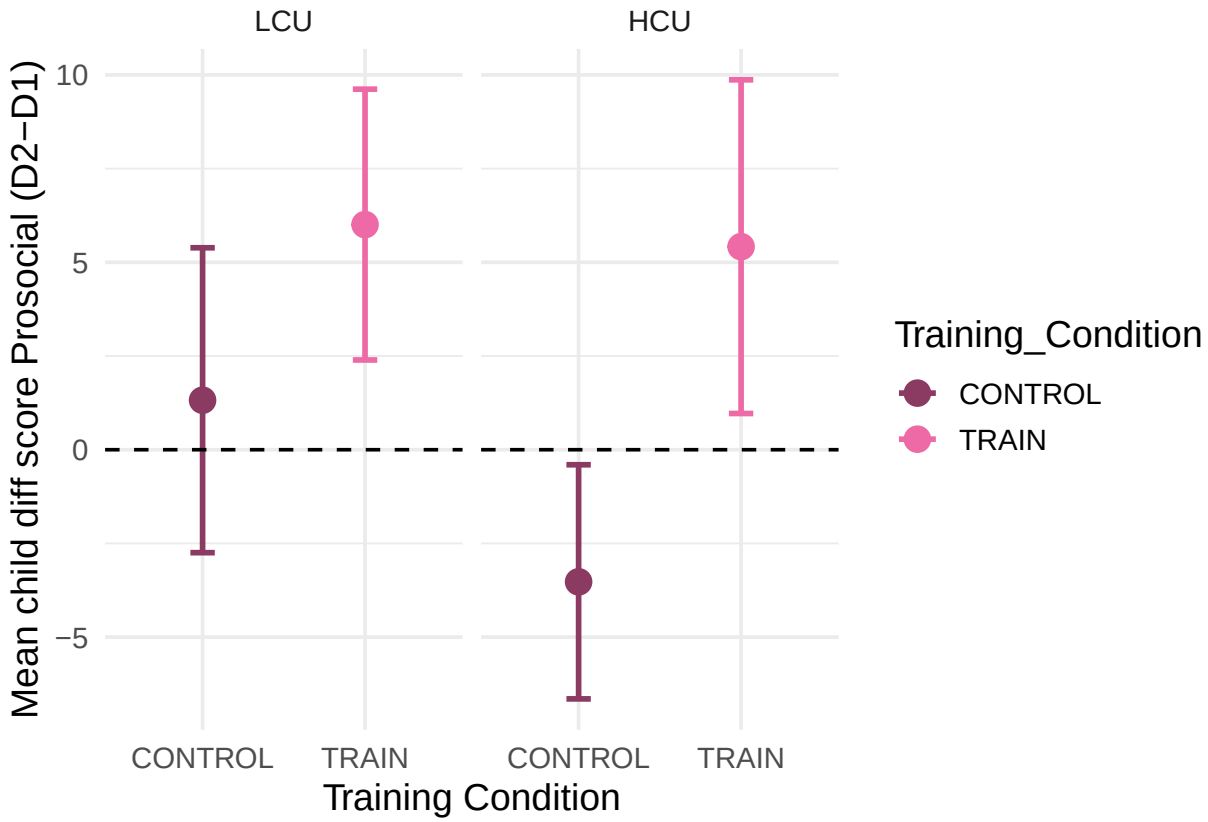
Training_Condition	diaryTP	CP_Group	Group_N_PBD	Mean_child_rated_PBD	se_child_rated_PBD	NA_child_rated_PBD
CONTROL	1	LCU	26	56.21923	3.738943	0
CONTROL	1	HCU	18	49.17444	4.593665	0
CONTROL	2	LCU	26	57.53923	4.655672	0
CONTROL	2	HCU	18	45.64944	5.016376	0
TRAIN	1	LCU	22	49.92909	5.136704	0
TRAIN	1	HCU	17	42.22706	5.347738	0
TRAIN	2	LCU	22	55.93545	7.295341	0
TRAIN	2	HCU	17	47.64471	6.633231	0

3.2) Plots of outcomes

First plotting data:



Difference plot:



3.3) Linear Regression

```
# singular fit so simplfying

model_c_psb_CU_lm <- lm(PSB_sum_D2 ~ PSB_sum_D1 + Training_Condition + CP_Group +
                          Training_Condition*CP_Group,
                        data =  child_psb_d1d2_wide)


# model without interaction
model_c_psb_CU_lm_nointeraction <- lm(PSB_sum_D2 ~ PSB_sum_D1 + Training_Condition + CP_Group,
                        data =  child_psb_d1d2_wide)
```

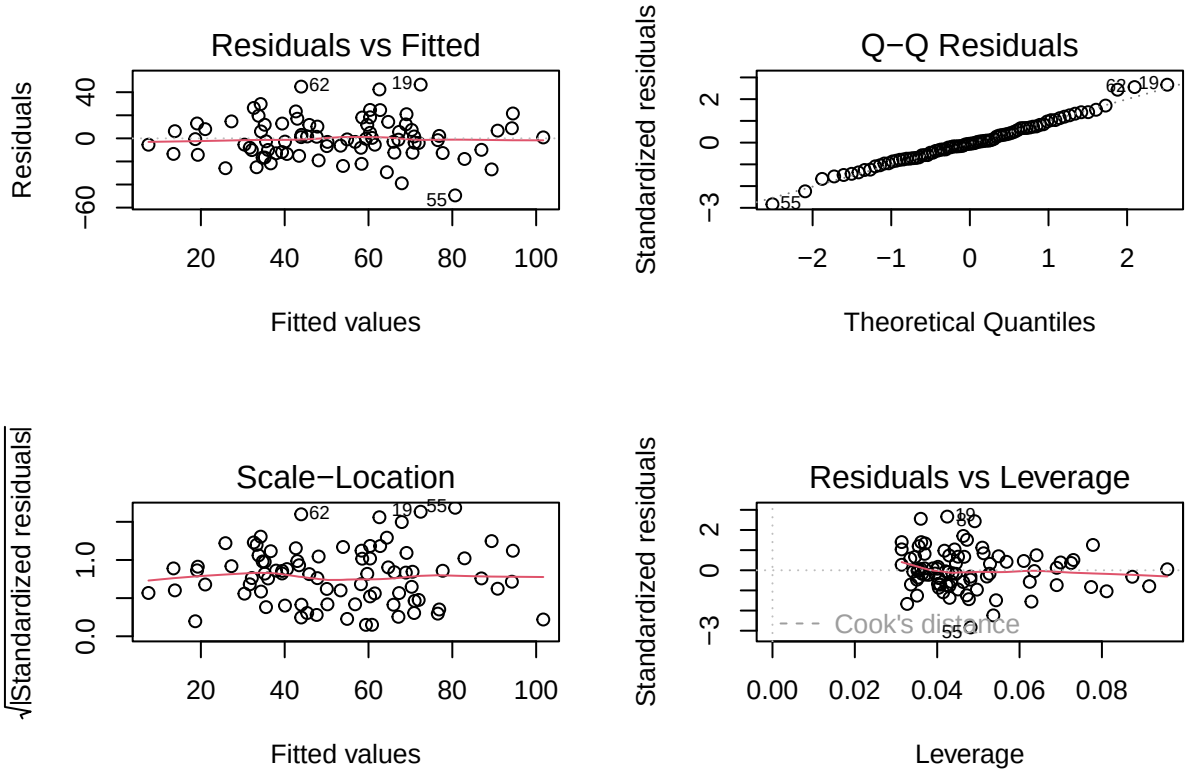
3.3.1) is model with interaction better fit? No

```
## Analysis of Variance Table
##
## Model 1: PSB_sum_D2 ~ PSB_sum_D1 + Training_Condition + CP_Group + Training_Condition *
##     CP_Group
## Model 2: PSB_sum_D2 ~ PSB_sum_D1 + Training_Condition + CP_Group
##   Res.Df   RSS Df Sum of Sq      F Pr(>F)
## 1     78 25123
## 2     79 25213 -1    -90.43 0.2808 0.5977
```

3.3.2) Model summary

```
##
## Call:
## lm(formula = PSB_sum_D2 ~ PSB_sum_D1 + Training_Condition + CP_Group,
##     data = child_psb_d1d2_wide)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -49.437 -12.119  -0.904  11.662  46.556
##
## Coefficients:
##                          Estimate Std. Error t value Pr(>|t|)
## (Intercept)                2.3638     6.1989   0.381    0.704
## PSB_sum_D1                 0.9668     0.0948  10.199 4.55e-16 ***
## Training_ConditionTRAIN    6.2697     3.9794   1.576    0.119
## CP_GroupHCU               -3.0710     4.0331  -0.761    0.449
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 17.86 on 79 degrees of freedom
## Multiple R-squared:  0.5833, Adjusted R-squared:  0.5675
## F-statistic: 36.87 on 3 and 79 DF,  p-value: 5.29e-15
```

3.3.3) Assumption checks



3.4) Bootstrapped linear regression

3.4.1) Results

```
boot_coef_c_psb <- function(data, indices) {
  d <- data[indices, ]  # Resample with replacement
  fit <- lm(PSB_sum_D2 ~ PSB_sum_D1 + Training_Condition + CP_Group,
                    data =  d)
  return(coef(fit))
}

# Run bootstrap
## Boot function: Generates R bootstrap replicates of a statistic applied to data.
## data = my data
## statistic = A function which, when applied to data returns a vector containing the statistic(s) of i
set.seed(123)  # For reproducibility
boot_results_c_psb <- boot(data = child_psb_d1d2_wide,
                      statistic = boot_coef_c_psb,
                      R = 5000)

# View results
boot_results_c_psb
```

```
##
## ORDINARY NONPARAMETRIC BOOTSTRAP
##
##
## Call:
## boot(data = child_psb_d1d2_wide, statistic = boot_coef_c_psb,
##     R = 5000)
##
##
## Bootstrap Statistics :
##       original       bias    std. error
## t1*  2.3637529  0.126875352  5.88209147
## t2*  0.9668249 -0.002304103  0.09138377
## t3*  6.2696675  0.045066967  3.90909055
## t4* -3.0710303 -0.064585294  3.91186123
```

3.4.2) Confidence Intervals

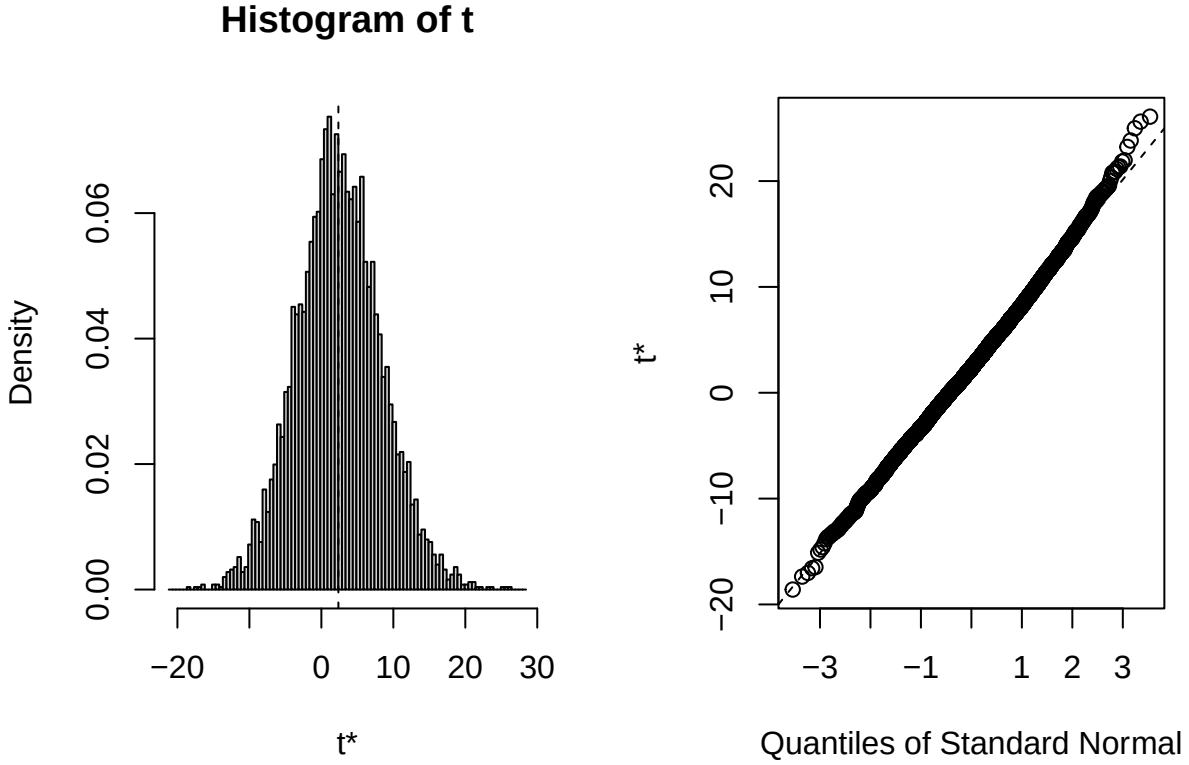
```
## BOOTSTRAP CONFIDENCE INTERVAL CALCULATIONS
## Based on 5000 bootstrap replicates
##
## CALL :
## boot.ci(boot.out = boot_results_c_psb, type = "bca", index = 1)
##
## Intervals :
## Level       BCa
## 95%   (-8.719, 14.785 )
## Calculations and Intervals on Original Scale
```

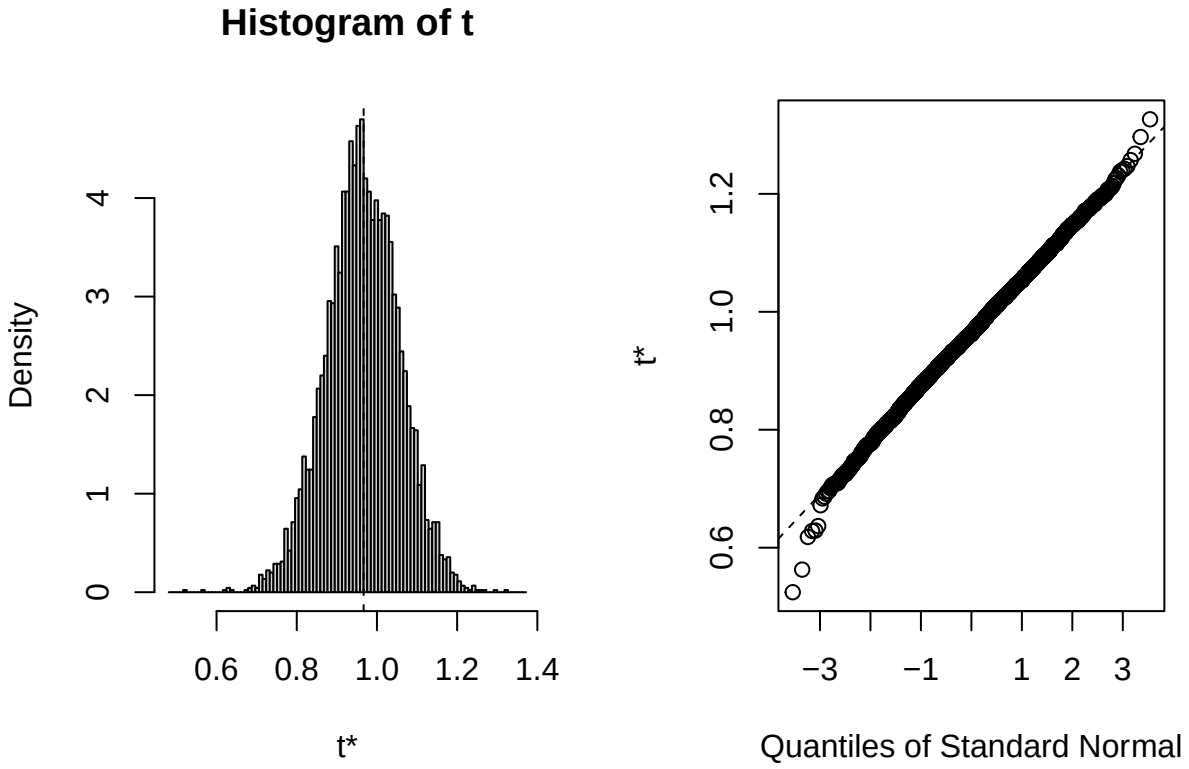
```
## BOOTSTRAP CONFIDENCE INTERVAL CALCULATIONS
## Based on 5000 bootstrap replicates
##
## CALL :
## boot.ci(boot.out = boot_results_c_psb, type = "bca", index = 2)
##
## Intervals :
## Level       BCa
## 95%   ( 0.7867,  1.1473 )
## Calculations and Intervals on Original Scale

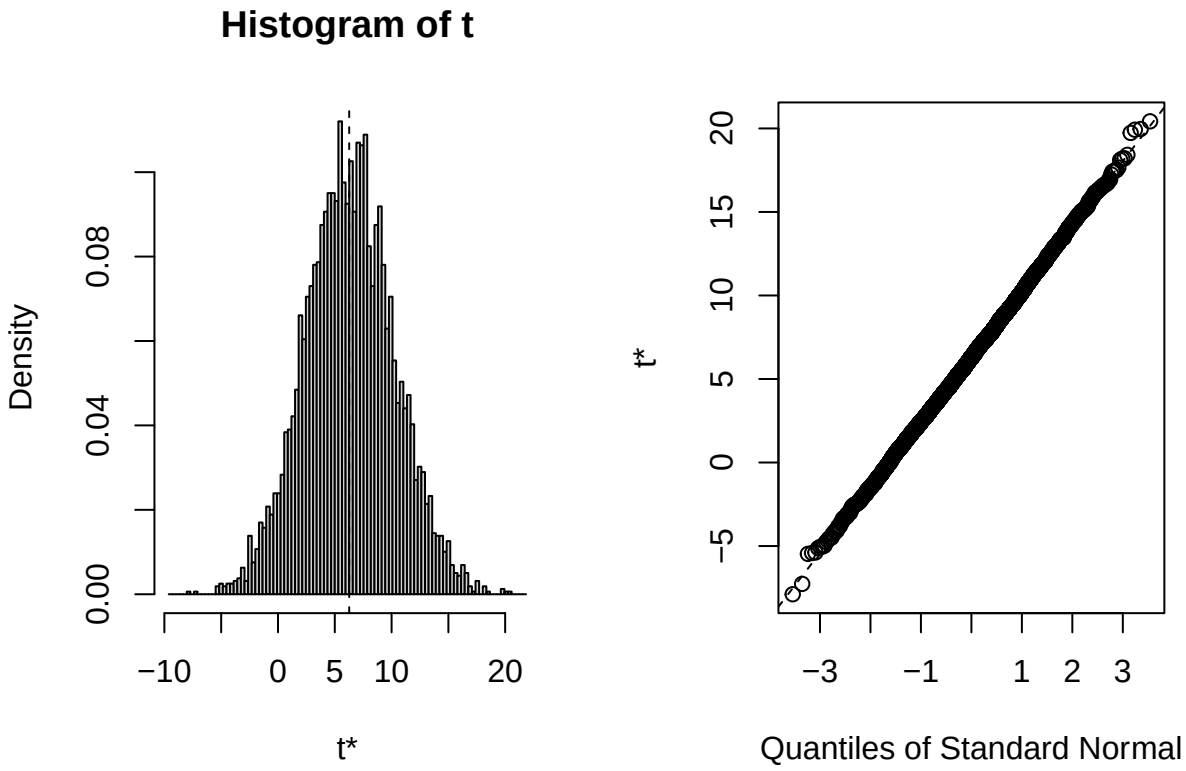
## BOOTSTRAP CONFIDENCE INTERVAL CALCULATIONS
## Based on 5000 bootstrap replicates
##
## CALL :
## boot.ci(boot.out = boot_results_c_psb, type = "bca", index = 3)
##
## Intervals :
## Level       BCa
## 95%   (-1.266, 14.274 )
## Calculations and Intervals on Original Scale

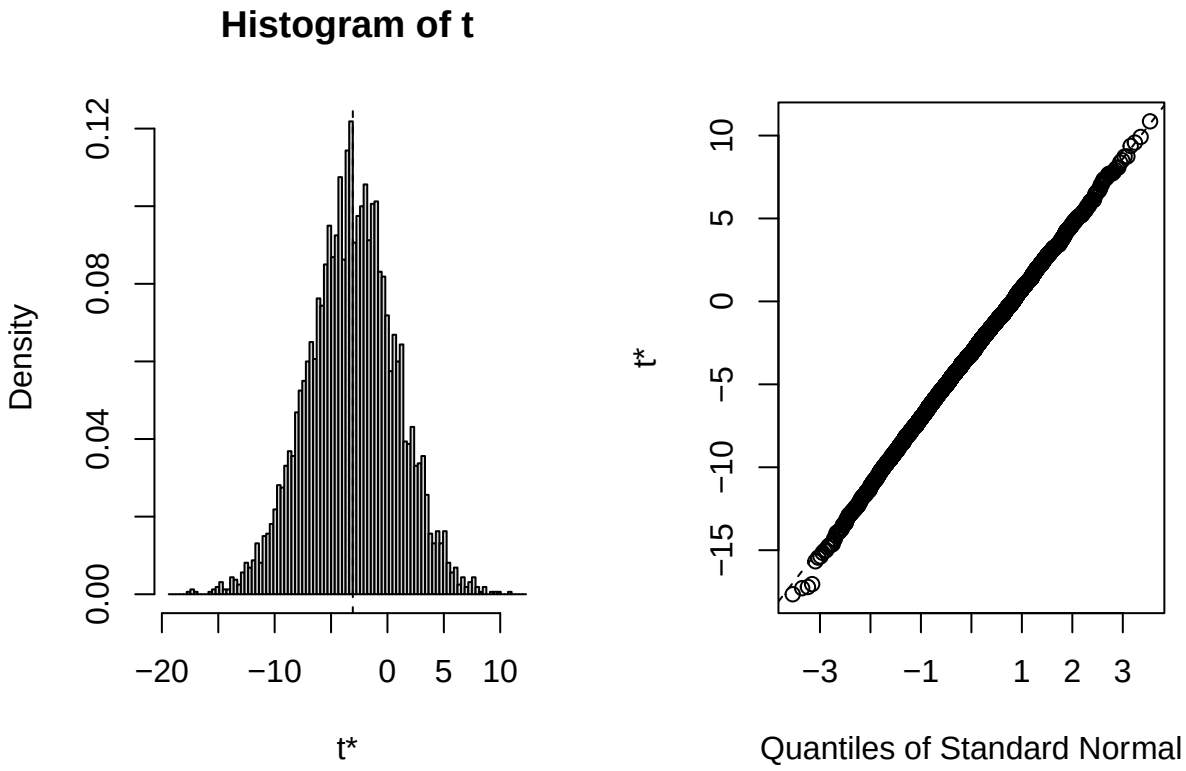
## BOOTSTRAP CONFIDENCE INTERVAL CALCULATIONS
## Based on 5000 bootstrap replicates
##
## CALL :
## boot.ci(boot.out = boot_results_c_psb, type = "bca", index = 4)
##
## Intervals :
## Level       BCa
## 95%   (-10.870,   4.565 )
## Calculations and Intervals on Original Scale
```

3.4.3) Model diagnostics






Histogram of t
Density
0.00
0.04
0.08
−10
0
5
10
20
t*
t*
−5
0
5
10
15
20
−3
−1
1
2
3
Quantiles of Standard Normal


Histogram of t
Density
0.00
0.04
0.08
0.12
−20
−10
0
5
10
t*
t*
10
5
0
−5
−10
−15
−3
−1
1
2
3
Quantiles of Standard Normal

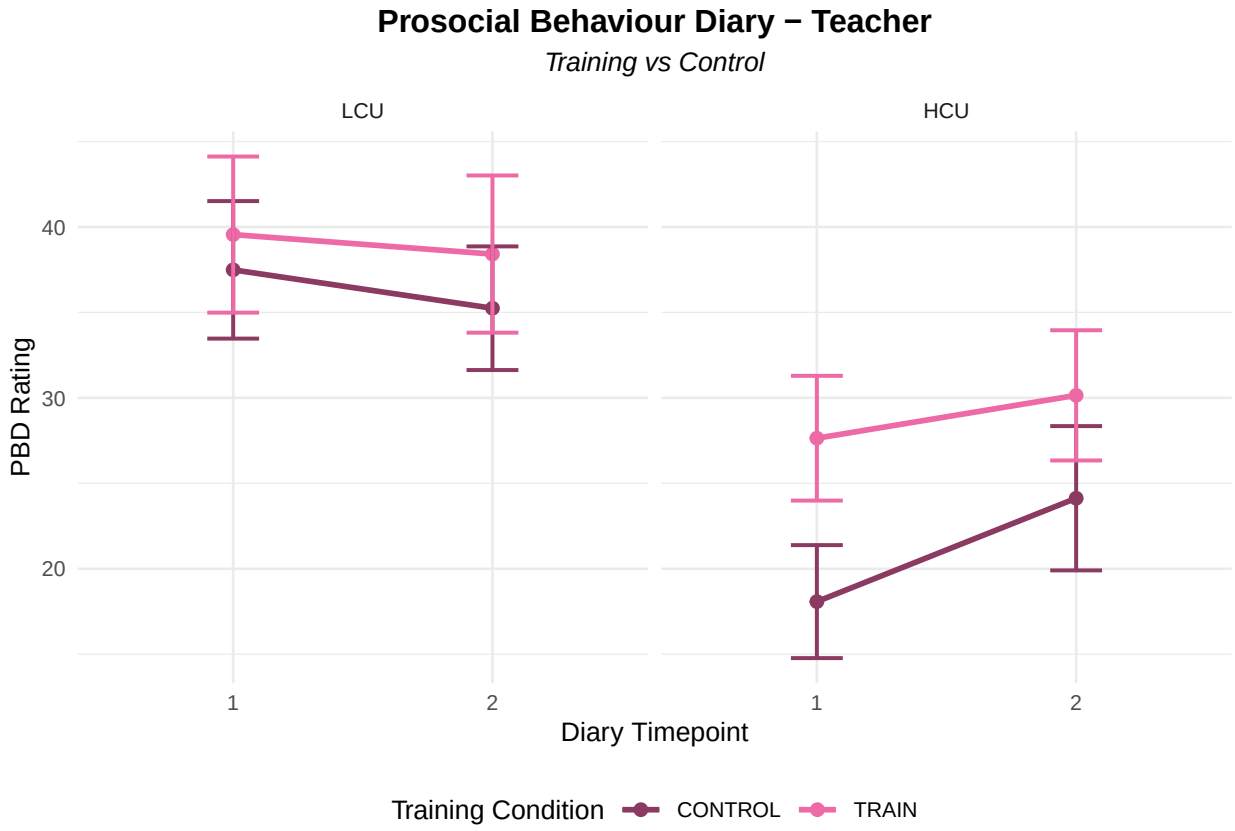
4) Teacher Prosocial at 2 weeks

4.1) Table of outcomes

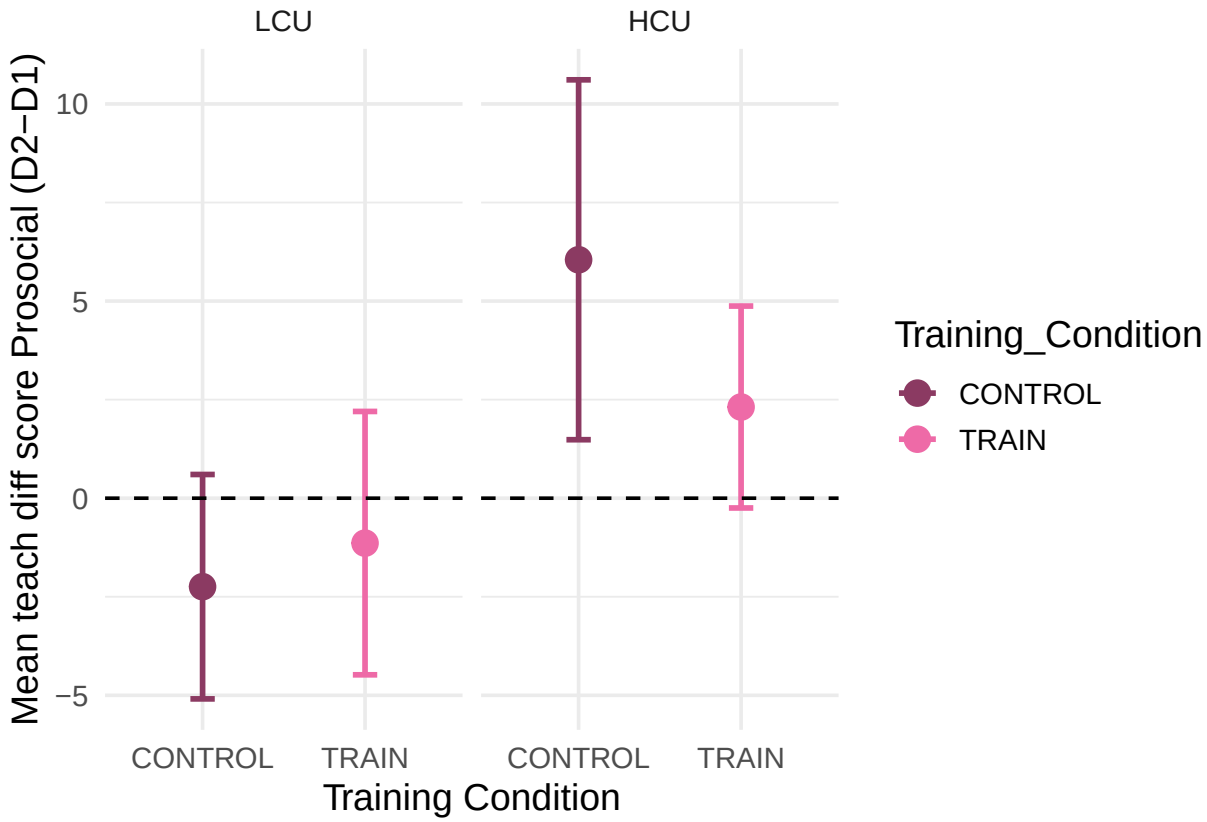
Training_Condition	diaryTP	CP_Group	Group_N	Mean_ABD	se_ABD	Mean_PBD	se_PBD
CONTROL	1	LCU	32	14.68229	3.379629	37.48958	4.023299
CONTROL	1	HCU	21	17.38889	3.301882	18.07937	3.305231
CONTROL	2	LCU	32	17.69792	3.158693	35.24479	3.616988
CONTROL	2	HCU	21	18.61905	3.537556	24.12698	4.220790
TRAIN	1	LCU	25	21.24000	4.115801	39.55333	4.566927
TRAIN	1	HCU	25	21.40000	3.943220	27.64000	3.650338
TRAIN	2	LCU	25	22.10000	2.898211	38.41333	4.598401
TRAIN	2	HCU	25	16.68000	3.256161	30.14667	3.809410

4.2) Plots of outcomes

First plotting data:



Difference plot:


LCU
HCU
Mean teach diff score Prosocial (D2−D1)
10
5
0
−5
CONTROL
TRAIN
CONTROL
TRAIN
Training Condition
Training_Condition
CONTROL
TRAIN

4.3) Linear Regression model

```
model_t_psb_CU_lm <- lm(PSB_mean_D2 ~ PSB_mean_D1 + Training_Condition + CP_Group +
                          Training_Condition*CP_Group,
                        data =  teach_psb_d1d2_wide)

# model without interaction
model_t_psb_CU_lm_nointeraction <- lm(PSB_mean_D2 ~ PSB_mean_D1 + Training_Condition + CP_Group,
                        data =  teach_psb_d1d2_wide)
```

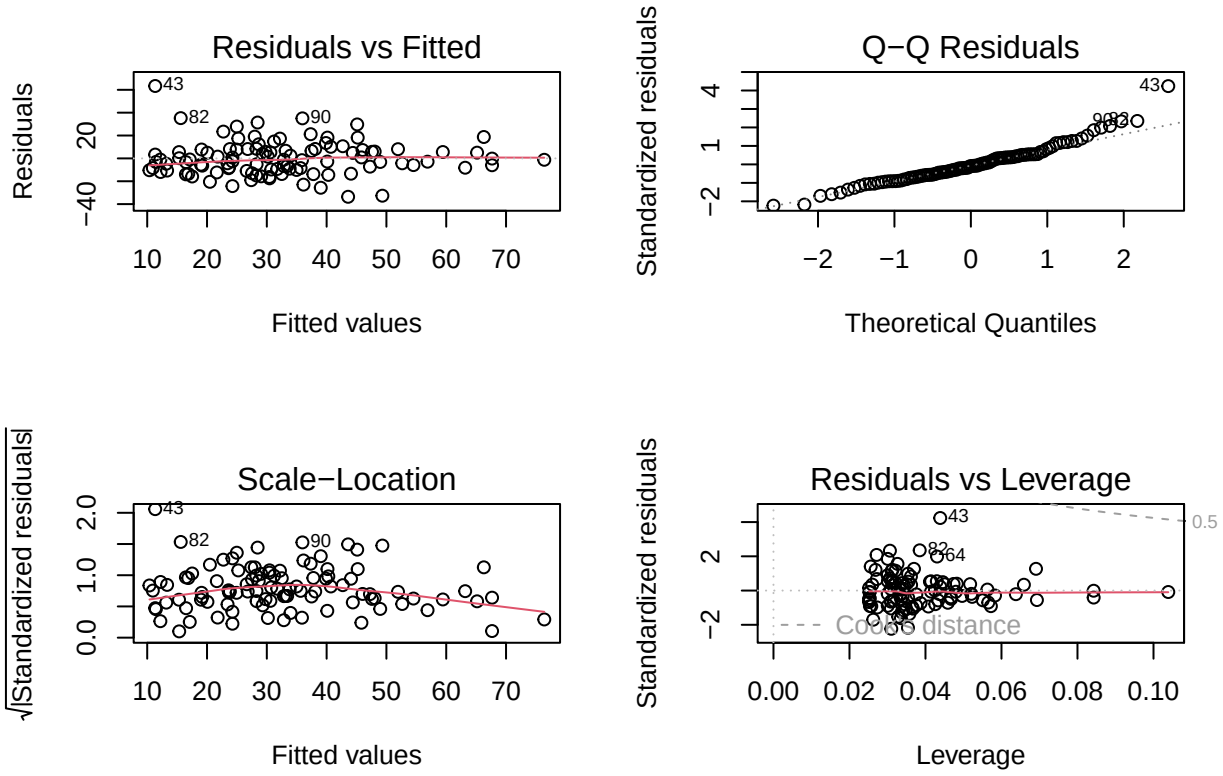
4.3.1) Does interaction term improve model? No

```
## Analysis of Variance Table
##
## Model 1: PSB_mean_D2 ~ PSB_mean_D1 + Training_Condition + CP_Group + Training_Condition *
##     CP_Group
## Model 2: PSB_mean_D2 ~ PSB_mean_D1 + Training_Condition + CP_Group
##   Res.Df   RSS Df Sum of Sq     F Pr(>F)
## 1     97 22866
## 2     98 22892 -1   -26.406 0.112 0.7386
```

4.3.2) Model results

```
{ r lm model results} summary(model_t_psb_CU_lm_nointeraction)
```

4.3.3) Assumption checks



4.4) Bootstrapped model

4.4.1) Results and specification

```
boot_coef_t_psb <- function(data, indices) {
  d <- data[indices, ]  # Resample with replacement
  fit <- lm(PSB_mean_D2 ~ PSB_mean_D1 + Training_Condition + CP_Group,
                    data =  d)
  return(coef(fit))
}

# Run bootstrap
## Boot function: Generates R bootstrap replicates of a statistic applied to data.
## data = my data
## statistic = A function which, when applied to data returns a vector containing the statistic(s) of i
set.seed(14423)  # For reproducibility
boot_results_t_psb <- boot(data = teach_psb_d1d2_wide,
                     statistic = boot_coef_t_psb,
                     R = 5000)

# View results
boot_results_t_psb
```

```
##
## ORDINARY NONPARAMETRIC BOOTSTRAP
##
##
## Call:
## boot(data = teach_psb_d1d2_wide, statistic = boot_coef_t_psb,
##     R = 5000)
##
##
## Bootstrap Statistics :
##       original        bias    std. error
## t1* 10.3765539  0.0319562598  3.67396567
## t2*  0.6739681 -0.0005688235  0.06336883
## t3*  0.8689605  0.0476087844  2.94561227
## t4*  0.9582111 -0.0531188909  3.06086605
```

4.4.2) Confidence Intervals

```
## BOOTSTRAP CONFIDENCE INTERVAL CALCULATIONS
## Based on 5000 bootstrap replicates
##
## CALL :
## boot.ci(boot.out = boot_results_t_psb, type = "bca", index = 1)
##
## Intervals :
## Level       BCa
## 95%   ( 4.10, 18.66 )
## Calculations and Intervals on Original Scale

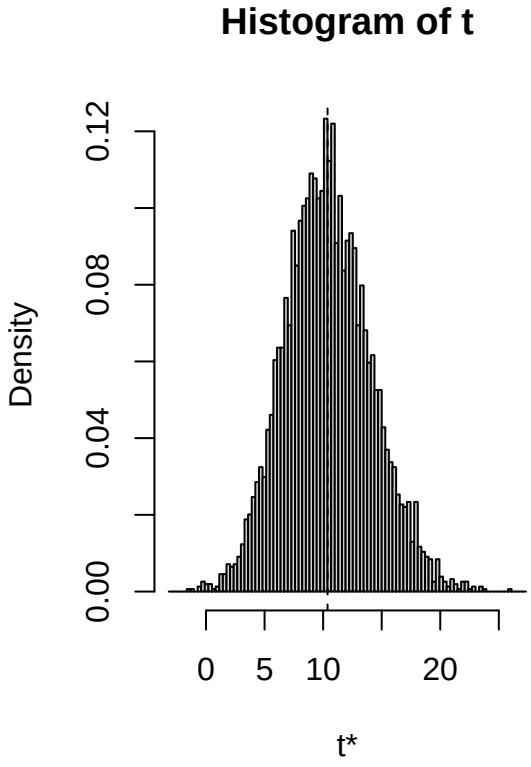
## BOOTSTRAP CONFIDENCE INTERVAL CALCULATIONS
## Based on 5000 bootstrap replicates
##
## CALL :
## boot.ci(boot.out = boot_results_t_psb, type = "bca", index = 2)
##
## Intervals :
## Level       BCa
## 95%   ( 0.5327,  0.7847 )
## Calculations and Intervals on Original Scale

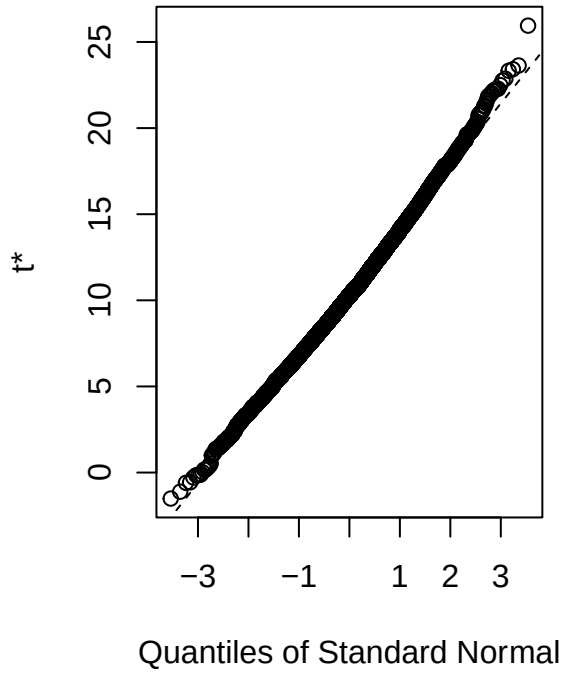
## BOOTSTRAP CONFIDENCE INTERVAL CALCULATIONS
## Based on 5000 bootstrap replicates
##
## CALL :
## boot.ci(boot.out = boot_results_t_psb, type = "bca", index = 3)
##
## Intervals :
## Level       BCa
## 95%   (-5.1278,  6.3304 )
## Calculations and Intervals on Original Scale

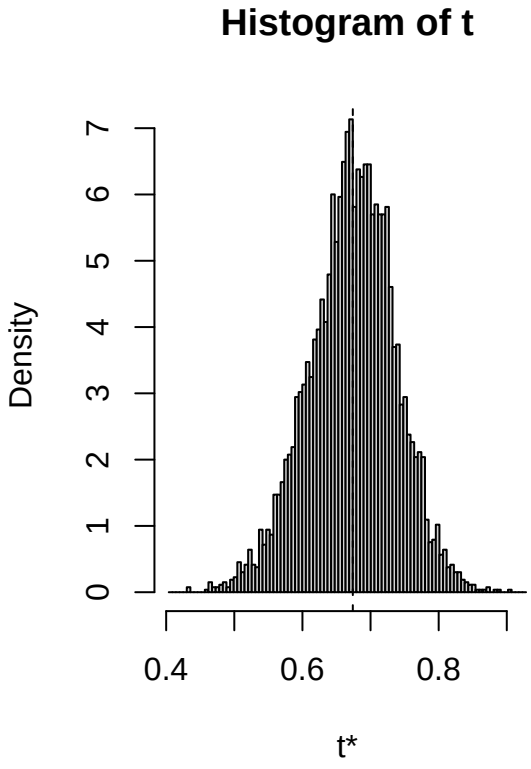
## BOOTSTRAP CONFIDENCE INTERVAL CALCULATIONS
## Based on 5000 bootstrap replicates
```

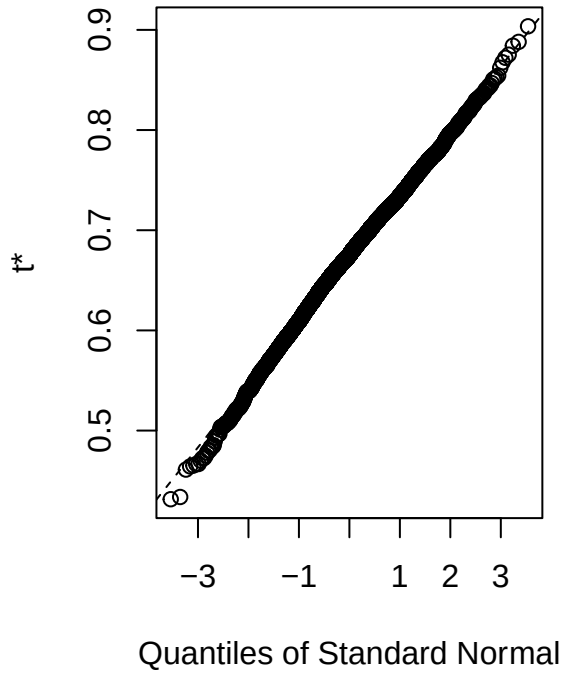
```
##
## CALL :
## boot.ci(boot.out = boot_results_t_psb, type = "bca", index = 4)
##
## Intervals :
## Level       BCa
## 95%   (-4.7618,  7.0929 )
## Calculations and Intervals on Original Scale
```

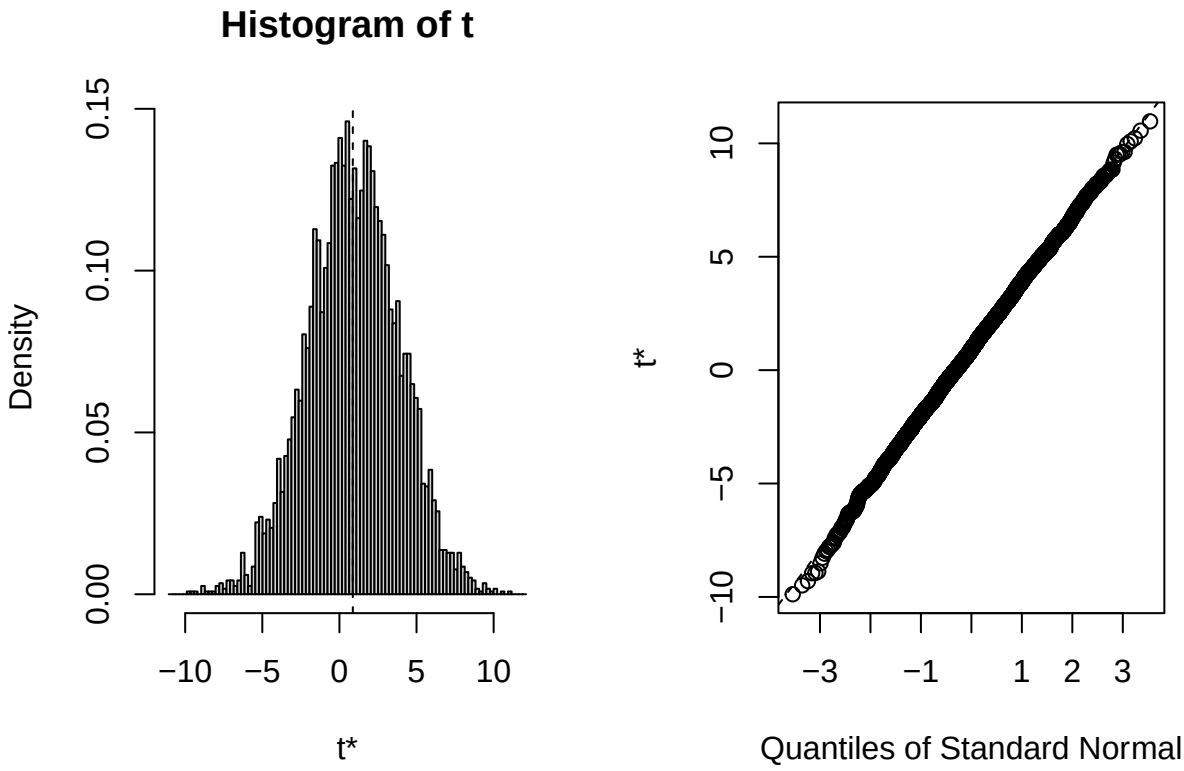
4.4.5) Model diagnostics

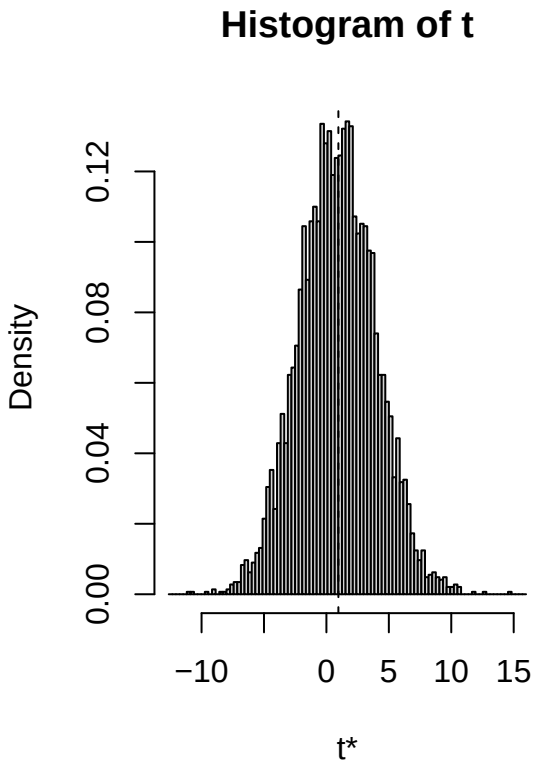


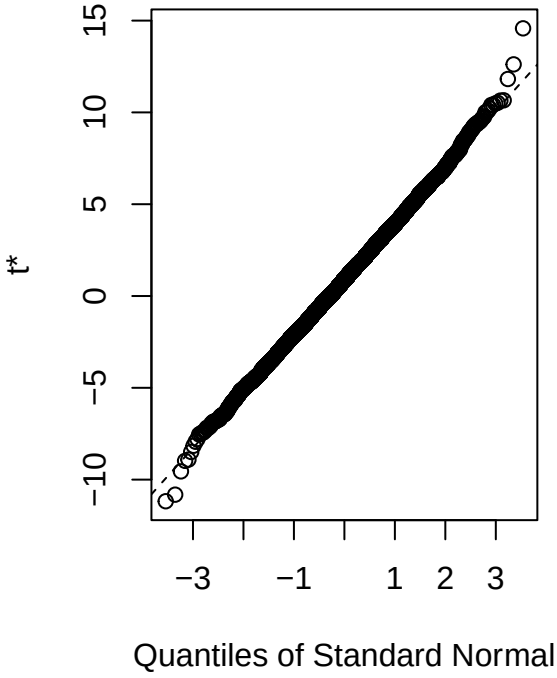







Histogram of t
Density
0.00
0.05
0.10
0.15
−10
−5
0
5
10
t*
t*
−10
−5
0
5
10
−3
−1
1
2
3
Quantiles of Standard Normal


Histogram of t
Density
0.00
0.04
0.08
0.12
−10
0
5
10
15
t*


t*
−10
−5
0
5
10
15
−3
−1
1
2
3
Quantiles of Standard Normal